Power



File: power.kicad_sch

Vehicle (OBD, CAN, K-Line)



File: vehicle.kicad_sch

Backup battery



File: battery.kicad_sch

MCU (ESP32-C5, USB)



File: mcu.kicad_sch

LTE + GNSS (EC200U)



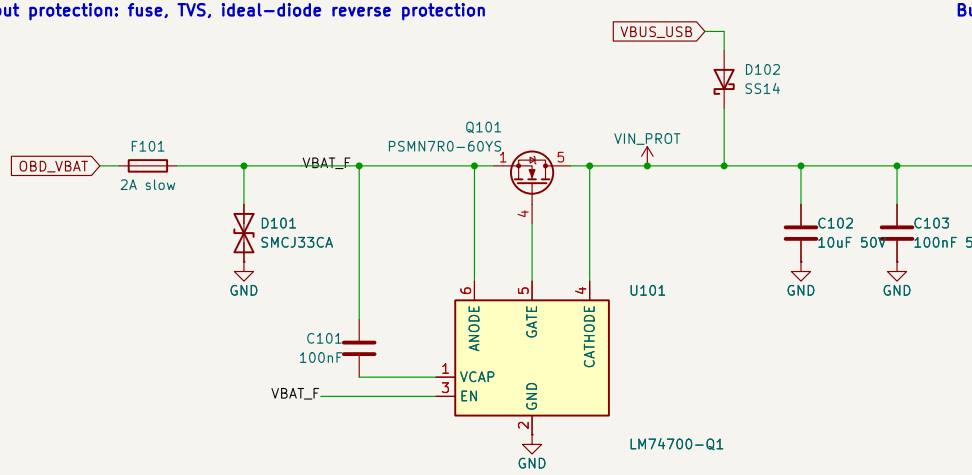
File: lte.kicad_sch

Sheet: /
File: obd-v2x-dongle.kicad_sch

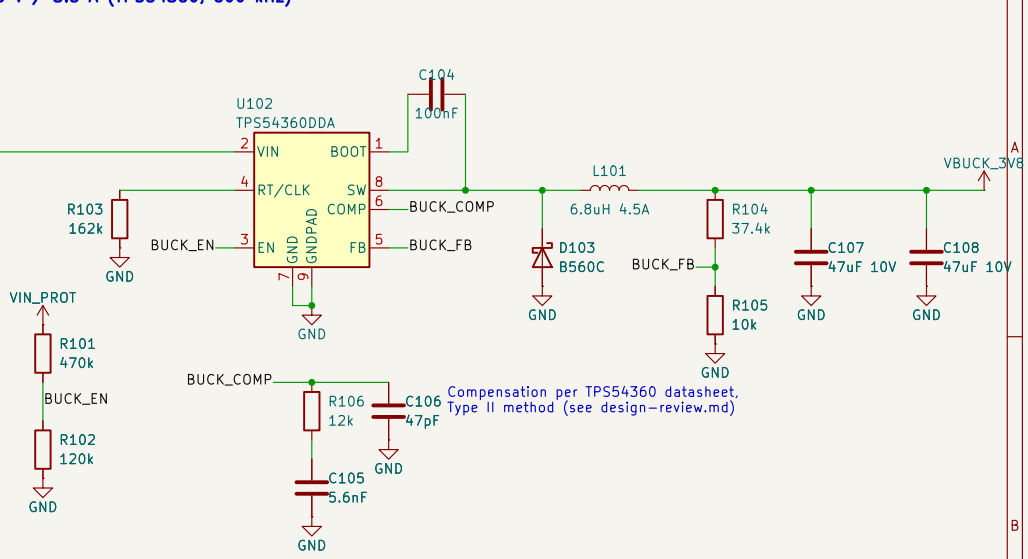
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 1/6

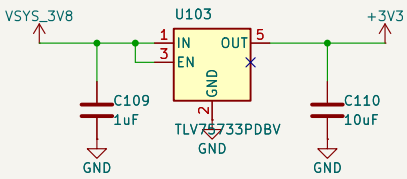
Input protection: fuse, TVS, ideal-diode reverse protection



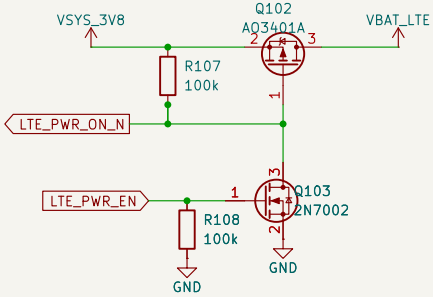
Buck 3.8 V / 3.5 A (TPS54360, 600 kHz)



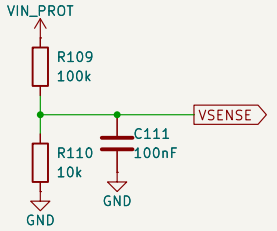
LDO 3.3 V (logic)



LTE modem supply switch (LTE_PWR_EN high = on)



Battery sense (VSENSE = VIN / 11, full scale ~35 V)



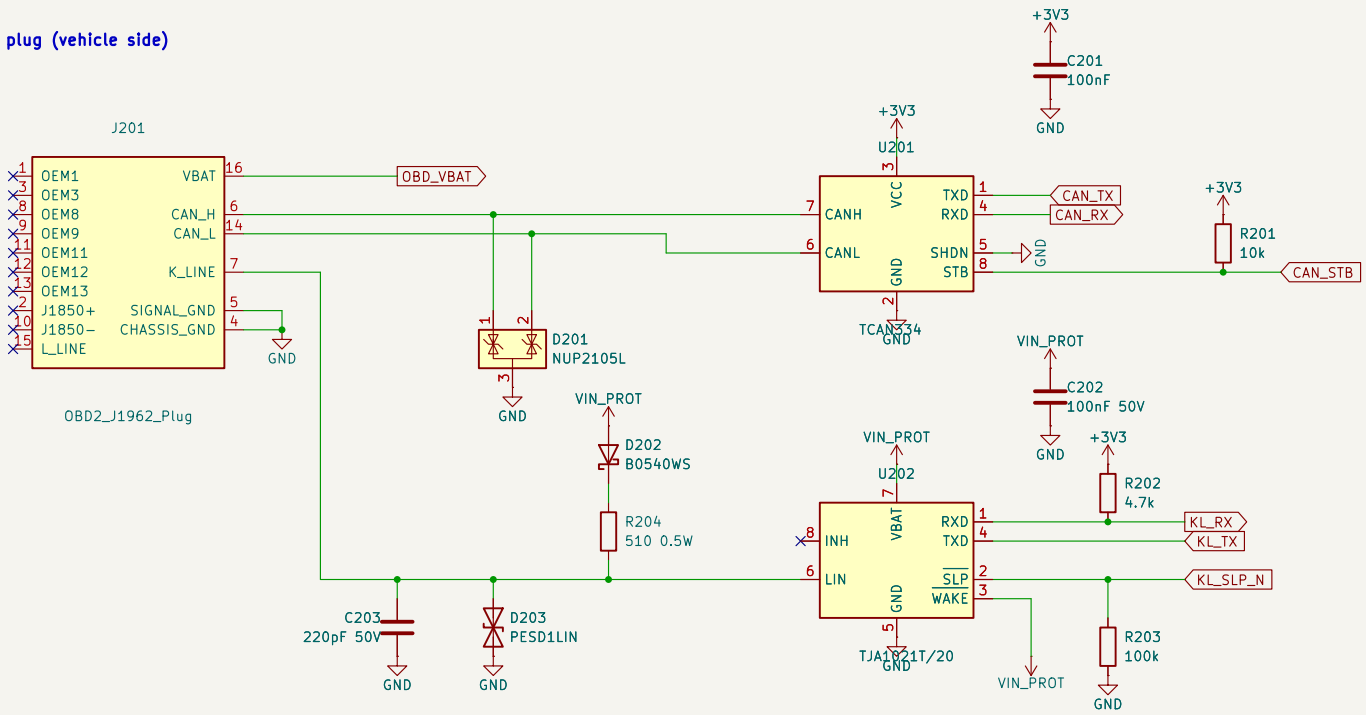
ERC power flags



Sheet: /Power/	
File: power.kicad_sch	
Title: Power: protection, buck 3.8 V, LDO 3.3 V	
Size: A4	Date:
KiCad E.D.A. 10.0.8	**Rev: 0.1**
	Id: 2/6

High-speed CAN (ISO 15765), no termination in the dongle

OBD-II plug (vehicle side)

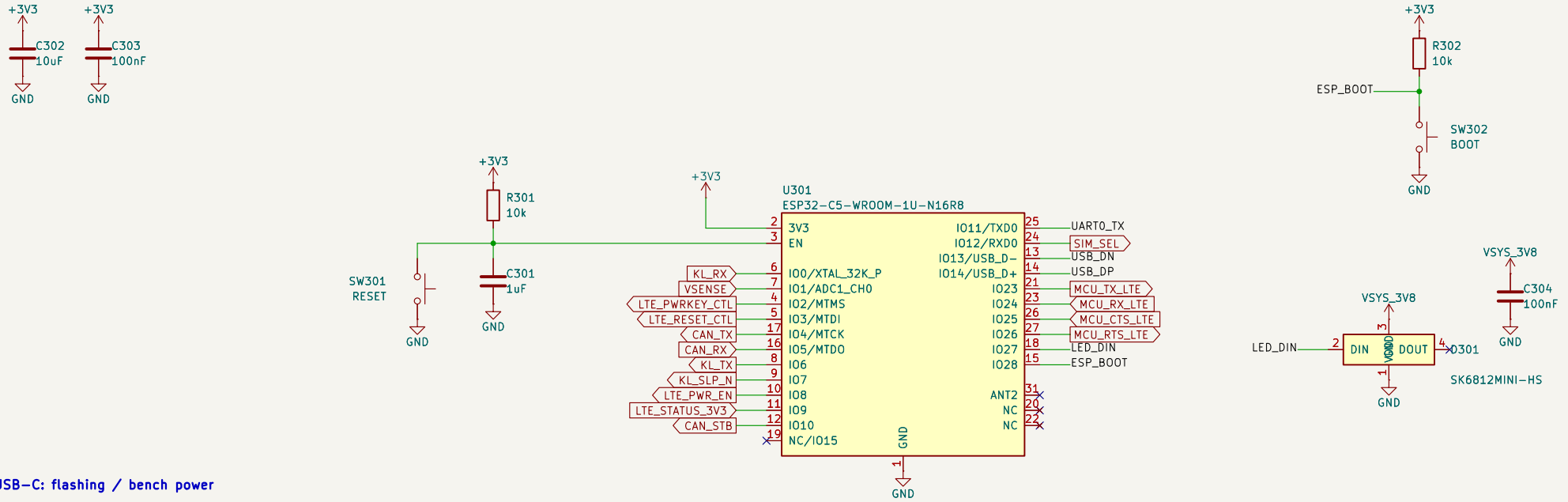


K-Line (ISO 9141 / KWP2000) via LIN transceiver + 510R tester pull-up

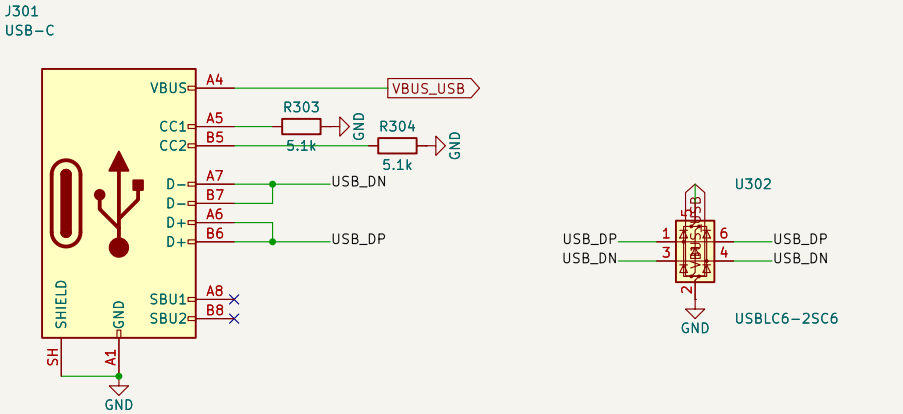
Sheet: /Vehicle (OBD, CAN, K-Line)/	
File: vehicle.kicad_sch	
Title: Vehicle interface: OBD-II, CAN, K-Line	
Size: A4	Date:
KiCad E.D.A. 10.0.8	Rev: 0.1
	Id: 3/6

ESP32-C5: 5.9 GHz ITS-G5 receive via U.FL on the module

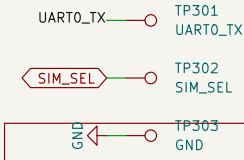
Boot / status LED



USB-C: flashing / bench power



Test pads: UART0 TX, SIM_SEL (IO12, was RXD0)



Sheet: /MCU (ESP32-C5, USB)/
File: mcu.kicad_sch

Title: MCU: ESP32-C5 (ITS-G5 RX, CAN FD, BLE), USB-C

Size: A4	Date:	Rev: 0.1
KiCad E.D.A. 10.0.8		Id: 4/6

SIM 2 = soldered eSIM U403 (fit an MFF2 eUICC)

PWRKEY / RESET_N open-collector drivers (MCU high = pull low)

LTE_PWRKEY_CTL
R401 4.7k
R402 47k
Q401 MMBT3904
LTE_PWRKEY
LTE_RESET_CTL
R403 4.7k
R404 47k
Q402 MMBT3904
LTE_RESET_N
GND

SIM2_VDD
C415 100nF
SIM2_RST R413 22R
SIM2_CLK R412 22R
SIM2_DAT R411 22R
U403 eSIM MFF2
VCC RST CLK I/O
GND VPP NC NC FP

1.8 V <-> 3.3 V level shifter (disabled while modem is off)

LTE_1V8
C405 100nF
+3V3
C406 100nF
U402 TXS0108EPW
OE VCCA VCCB GND
A1 A2 A3 A4 A5 A6 A7 A8
B1 B2 B3 B4 B5 B6 B7 B8
LTE_RXD_1V8
LTE_TXD_1V8
LTE_RTS_1V8
LTE_CTS_1V8
LTE_STATUS
MCU_TX_LTE
MCU_RX_LTE
MCU_RTS_LTE
MCU_CTS_LTE
LTE_STATUS_3V3

+3V3
C420 100nF
U405 TPS2552DBVR-1
IN EN ILIM GND OUT FAULT
LTE_PWR_ON_N
R415 232k
C419 100nF
C418 33pF
L401 27nH
LTE_ANT_GNSS_UFL

GNSS antenna bias (active + passive antennas)
3.3 V while the modem is on; latch-off 100–137 mA

U401 EC200U-EU
VBAT_BB VBAT_BB VBAT_RF VBAT_RF VDD_EXT
PWRKEY RESET_N STATUS NET_STATUS NET_MODE W_DISABLE# AP_READY WAKEUP_IN USB_BOOT
USB_VBUS USB_DP USB_DM
ADC0 ADC1
RXD TXD RTS CTS DTR RI DCD DBG_RXD DBG_TXD
USIM_VDD USIM_DATA USIM_CLK USIM_RST USIM_DET USIM_GND
ANT_MAIN ANT_BT/WIFI ANT_GNSS
LTE_RXD_1V8
LTE_TXD_1V8
LTE_RTS_1V8
LTE_CTS_1V8
LTE_PWRKEY
LTE_RESET_N
LTE_STATUS
LTE_USB_BOOT
LTE_USB_VBUS
LTE_USB_DP
LTE_USB_DM
LTE_DBG_RXD
LTE_DBG_TXD
SIM_VDD
LTE_USIM_DATA
LTE_USIM_CLK
LTE_USIM_RST
LTE_ANT_MAIN
LTE_ANT_GNSS

Test pads: modem USB
(firmware update), debug UART

TP401 LTE_USB_VBUS
TP402 LTE_USB_DP
TP403 LTE_USB_DM
TP404 LTE_USB_BOOT
TP405 LTE_DBG_TXD
TP406 LTE_DBG_RXD
TP407 LTE_1V8

SIM 1 = nano-SIM J401 (default)

SIM1_RST R407 22R
SIM1_CLK R406 22R
SIM1_DAT R405 22R
J401 Nano-SIM
SIM1_VDD SIM_RST SIM_CLK GND VPP SIM_IO

VBAT_LTE bulk/decoupling (close to pins 57–60)

C401 220uF 6.3V
C402 100nF
C403 33pF
C404 10pF

+3V3
C416 100nF

U404 FSA2567UMX
VCC VSIM RST CLK DAT Sel GND NC
1VSIM 1RST 1CLK 1DAT 2VSIM 2RST 2CLK 2DAT
SIM1_VDD SIM1_RST SIM1_CLK SIM1_DAT SIM2_VDD SIM2_RST SIM2_CLK SIM2_DAT
SIM_SEL
R414 100k

C407 33pF
C408 33pF
C409 33pF
C410 100nF
R408 15k

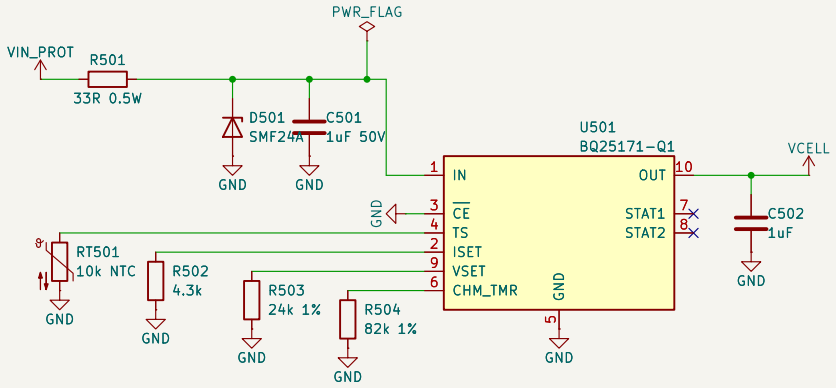
SIM switch: SIM_SEL low (default) = nano-SIM J401, high = eSIM U403

Antennas (pi-match, shunt caps DNP until tuned)

J402 U.FL LTE main
J403 U.FL GNSS
LTE_ANT_MAIN
LTE_ANT_GNSS
R409 0R
C411 DNP
C412 DNP
C417 33pF
C413 DNP
C414 DNP

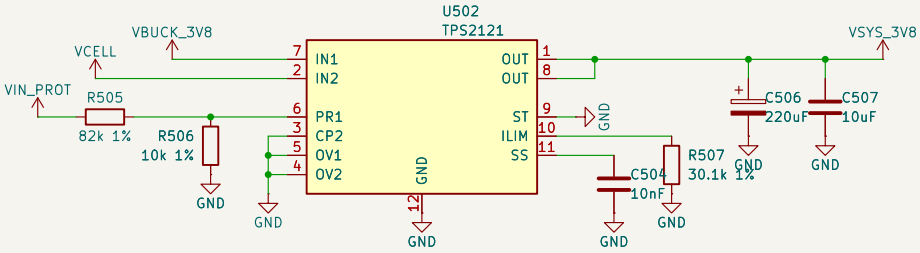
Sheet: /LTE + GNSS (EC200U)/		
File: lte.kicad_sch		
Title: LTE Cat 1 + GNSS: Quectel EC200U-EU (1.8 V I/O)		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 5/6

Li-ion charger, fed from the car side (VIN_PROT)



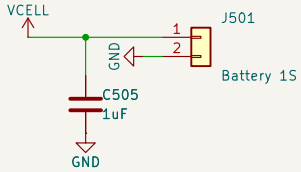
ISET 4.3k: 70 mA charge (kept low: linear from 14 V, ~0.6 W)
VSET 24k: 4.10 V float CHM_TMR 82k: Li-ion, 10 h timer
TS: 10k NTC next to the cell -> charging only 0..45 C

Priority power mux: buck (IN1) while VIN_PROT > ~9.7 V, else cell (IN2)



ILIM 30.1k: ~3.5 A SS 10 nF
Firmware on battery: LTE only (lower burst current), shut the modem down below ~3.6 V cell

1S Li-ion/LiPo cell, 250-400 mAh, with protection PCM



Sheet: /Backup battery/
File: battery.kicad_sch

Title: Backup battery: 1S Li-ion charger (BQ25171-Q1) + priority power mu

Size: A4	Date:	Rev: 0.1
KiCad E.D.A. 10.0.8		Id: 6/6